

Quantum Mechanics

Science Team

SoCSE, RV University

December 16, 2024



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TIMELINE OF QUANTUM MECHANICS

MILESTONES

- 1801 ● Young's Double-Slit Experiment
- 1802 ● Dark lines in the solar spectrum
- 1814 ● Fraunhofer lines
- 1859 ● Emission Spectra
- 1887 ● Discovery of Photoelectric Effect
- 1900 ● Black body Radiation
- 1900 ● Planck's Law
- 1905 ● Einstein's Photoelectric Equation
- 1913 ● Bohr's Model
- 1924 ● de Broglie Wavelength
- 1925 ● Heisenberg's Uncertainty
- 1926 ● Schrödinger Equation
- 1930s ● Postulates of Quantum Mechanics
- 1935 ● Superposition and Entanglement

Young's Double-Slit Experiment

Thomas Young's experiment demonstrated the wave nature of light by producing an interference pattern through two slits.



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Dark lines in the solar spectrum

William Hyde Wollaston discovered dark lines observed in the solar spectrum using a prism.



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Fraunhofer lines

Fraunhofer lines are the dark lines in the solar spectrum and correspond to specific wavelengths absorbed by elements in the Sun's atmosphere.



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Emission Spectra

Gustav Kirchhoff and Robert Bunsen discovered that atoms or molecules emit unique wavelengths of light emitted by atoms or molecules when they are excited.



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Discovery of Photoelectric Effect

The photoelectric effect was discovered when light striking a metal surface caused the emission of electrons, observed by Heinrich Hertz.



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Black body Radiation

The electromagnetic radiation emitted by a perfect absorber and emitter of energy, with its spectrum depending only on the temperature of the object.



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Planck's Law

Planck's Law describes the distribution of electromagnetic radiation emitted by a blackbody, introducing the idea that energy is quantized.



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Einstein's Photoelectric Equation

Einstein's photoelectric equation explains the photoelectric effect by relating the kinetic energy of emitted electrons to the frequency of incoming light and the work function of the material.



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Bohr's Model

Bohr's model of the atom introduced quantized electron orbits to explain the discrete spectral lines observed in hydrogen's emission spectrum.



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de Broglie Wavelength

Louis de Broglie proposed that particles, like electrons, have wave-like properties, with a wavelength inversely proportional to their momentum.



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Heisenberg's Uncertainty

Heisenberg's uncertainty principle asserts that it is impossible to simultaneously measure both the position and momentum of a particle with arbitrary precision.



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Schrödinger Equation

Schrödinger's equation describes how the quantum state of a physical system evolves over time, using the wavefunction to predict probabilities of outcomes.



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Postulates of Quantum Mechanics

The postulates of quantum mechanics form the foundation of the theory, stating that physical systems are described by wavefunctions and measurements are probabilistic.



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Superposition and Entanglement

Superposition allows a quantum system to exist in multiple states simultaneously, while entanglement describes a correlation between particles that cannot be separated, even at a distance.



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EVENTS LEADING UP TO YOUNG'S DOUBLE SLIT EXPERIMENT

Proposed light as a particle-like entity

Pythagoras proposed light as a particle-like entity and believed it was connected to geometry and harmony in the universe.

Proposed light as a particle-like entity		circa 500 BCE
Physical nature of light		circa 400 BCE
Development of geometrical optics		circa 300 BCE
Camera Obscura and Color theory		965–1040 CE
Law of Refraction		1637
Fermat's Principle of Least Time		1657–1662
Early wave theory of light		1665
Wave theory of light		1678
Corpuscular theory of light		1704
Young's Double Slit Experiment		1801



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Physical nature of light

Empedocles and Plato theorized that light had a physical nature, with early ideas suggesting that light was somehow emitted from the eye or from objects.

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Development of geometrical optics

In his work *Optica*, he explored the behavior of light, particularly reflection, and stated that light travels in straight lines.

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Physical nature of light	●	circa 400 BCE
Development of geometrical optics	●	circa 300 BCE
Camera Obscura and Color theory	●	965–1040 CE
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Camera Obscura and Color theory

Ibn al-Haytham's ground breaking contributions, including but not limited to reflection, refraction, camera obscura, color theory, and much more

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Law of Refraction

In Dioptrique, Descartes introduced the law of refraction and proposed light as a physical disturbance or vibration in a medium.

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Fermat's Principle of Least Time

Fermat formulated the principle that light travels between two points along the path that requires the least time.

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EVENTS LEADING UP TO YOUNG'S DOUBLE SLIT EXPERIMENT

Early wave theory of light

Robert Hooke observed diffraction and developed some early ideas about the wave nature of light.

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EVENTS LEADING UP TO YOUNG'S DOUBLE SLIT EXPERIMENT

Wave theory of light

Christiaan Huygens suggested the wave theory of light, explaining diffraction and interference phenomena.

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Corpuscular theory of light

Proposed the particle theory of light, suggesting light was composed of small particles (corpuscles), and used this theory to explain reflection, refraction, and color.

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EVENTS LEADING UP TO YOUNG'S DOUBLE SLIT EXPERIMENT

Young's Double Slit Experiment

Young's double-slit experiment in 1801 finally provided clear experimental evidence for the wave nature of light by demonstrating interference patterns, a phenomenon that could not be explained by the particle theory.

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SO THE QUESTION IS

What is Light?

SO THE QUESTION IS

What is Light?

A Wave?

SO THE QUESTION IS

What is Light?

A Wave?

or

SO THE QUESTION IS

What is Light?

A Wave?

or

A Particle?

SO THE QUESTION IS

What is wave-particle duality?

SO THE QUESTION IS

Why did Einstein win the Nobel?

SO THE QUESTION IS

How did all the developments lead to the
Nobel for Quantum Computing in 2022?

SO THE QUESTION IS

What is Quantum Mechanics?

YOUNG'S DOUBLE SLIT EXPERIMENT



Figure: Diffraction pattern of Young's Double Slit Experiment



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YOUNG'S DOUBLE SLIT EXPERIMENT

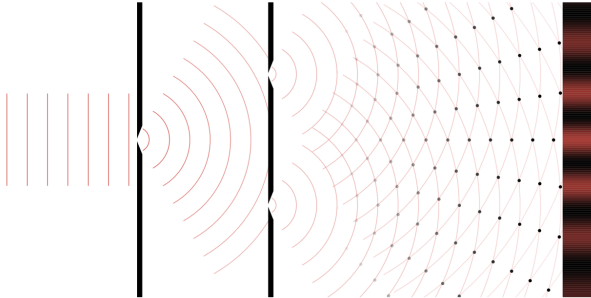


Figure: Schematic of a Young's Double Slit Experiment

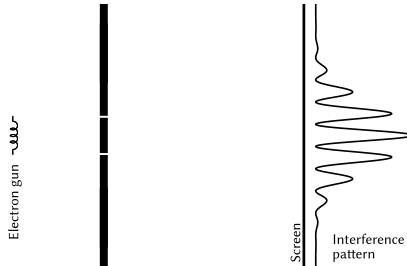
https://javalab.org/en/youngs_double_slit_en/

YOUNG'S DOUBLE SLIT EXPERIMENT

- NOW WITH ELECTRONS

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- NOW WITH ELECTRONS



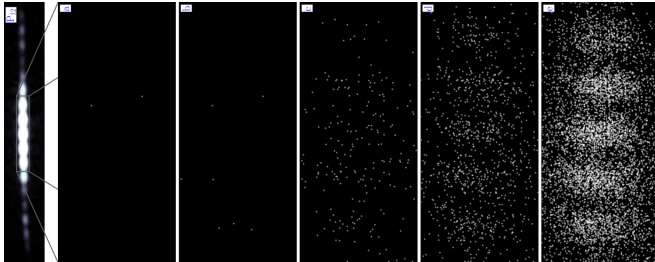
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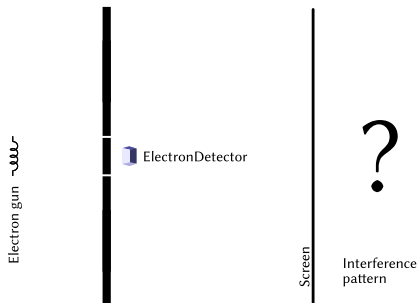
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YOUNG'S DOUBLE SLIT EXPERIMENT

- THE WHICH-WAY EXPERIMENT

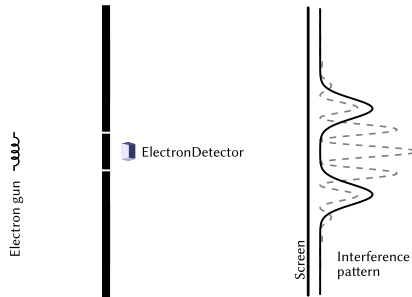
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YOUNG'S DOUBLE SLIT EXPERIMENT

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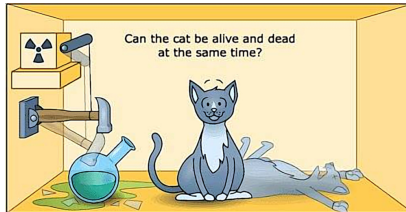
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MEASUREMENT - MEASURE ONLY WHEN YOU'RE READY

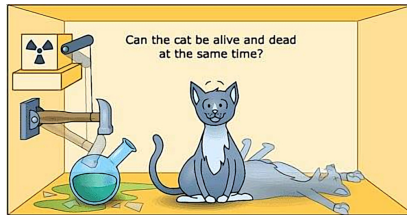
MEASUREMENT - MEASURE ONLY WHEN YOU'RE READY

**Measurement destroys the state of the
system!**

SUPERPOSITION - IT'S LIKE BEING AT TWO PLACES AT ONCE



SUPERPOSITION - IT'S LIKE BEING AT TWO PLACES AT ONCE



The cat is both dead and alive at the same time, until you measure!

YOUNG'S DOUBLE SLIT EXPERIMENT

- WHICH WAY?

YOUNG'S DOUBLE SLIT EXPERIMENT

- WHICH WAY?

**The electron (or a photon) goes both ways
until you try to measure!**

YOUNG'S DOUBLE SLIT EXPERIMENT

- SUMMARY

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- 1 **Classic Experiment Overview:** Thomas Young conducted his double slit experiment in 1801 to demonstrate the wave nature of light.

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- 1 **Classic Experiment Overview:** Thomas Young conducted his double slit experiment in 1801 to demonstrate the wave nature of light.
- 2 **Single Electron Variant:** Even when electrons are sent one by one, an interference pattern still emerges over time highlighting it's wave nature.

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- ➋ **Single Electron Variant:** Even when electrons are sent one by one, an interference pattern still emerges over time highlighting it's wave nature.
- ➌ **Role of Measurement:** The act of measuring which slit an electron (or photon) passes through collapses its wave function into a definite state (particle-like behavior).

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- ➍ **Principle of Superposition:** The principle of superposition states that in quantum mechanics, a particle exists in multiple states simultaneously until measured.
- ➎ **Wave-Particle Duality:** Young's experiment exemplifies the concept of wave-particle duality in quantum mechanics.

SIR ISAAC NEWTON AND THE COMPOSITION OF SUNLIGHT

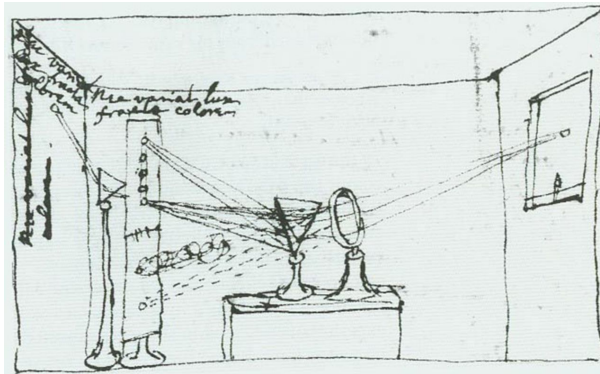


Figure: Newton's sketch of his experiment. Courtesy: Royal Society



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FRAUNHOFER LINES



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FRAUNHOFER LINES

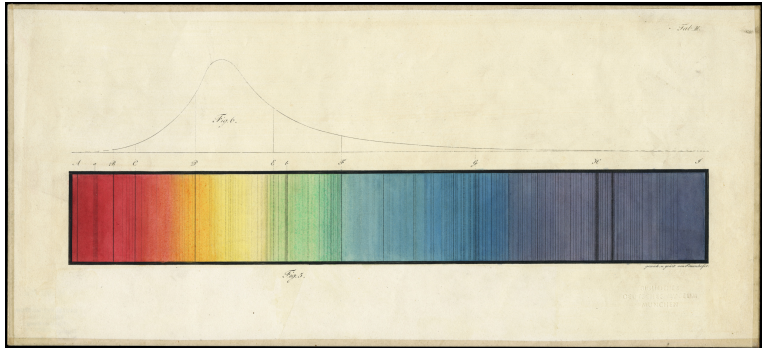
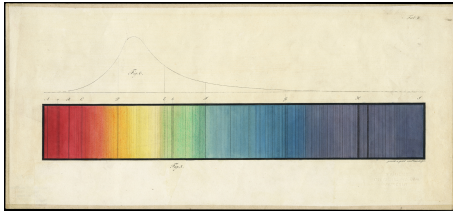


Figure: Solar spectrum drawn and colored by Joseph von Fraunhofer. Small script in the lower right corner reads "gezeich. u. geätzt von Fraunhofer" Courtesy of Deutsches Museum

FRAUNHOFER LINES



Although William Hyde was the first person to discover the dark lines of solar spectrum, it was Joseph von Fraunhofer who later rediscovered and systematically studied these dark lines in 1814.

KIRCHHOFF, BUNSEN AND THE EMISSION SPECTRA

- Gustav Kirchhoff noted dark lines from a source giving a continuous spectrum when passed through salt flames - Fraunhofer D lines.

KIRCHHOFF, BUNSEN AND THE EMISSION SPECTRA

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- Identified the emission spectra of the salt - it corresponded to the dark lines!

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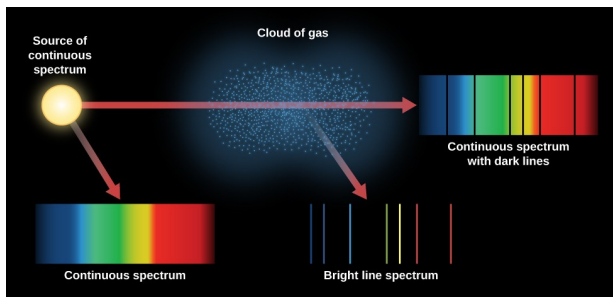
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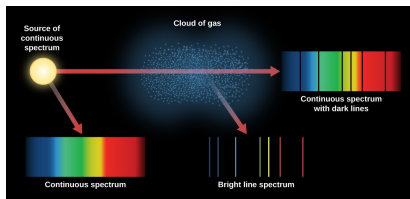
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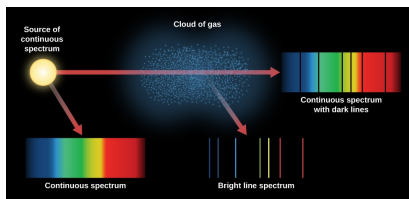
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A hot solid, liquid, or dense gas produces a continuous spectrum.

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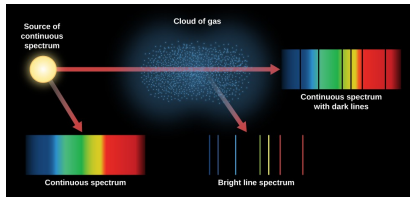
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A cool, low-density gas in front of a hotter source of a continuous spectrum produces an absorption line spectrum.

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These are empirical laws - laws given purely based on experimental observations.

SPECTRAL LINES, RYDBERG AND HIS FORMULA

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$R_H = 1.09737 \times 10^7 \text{ m}^{-1}$ is the Rydberg constant.

THE BLACK BODY AND IT'S RADIATION

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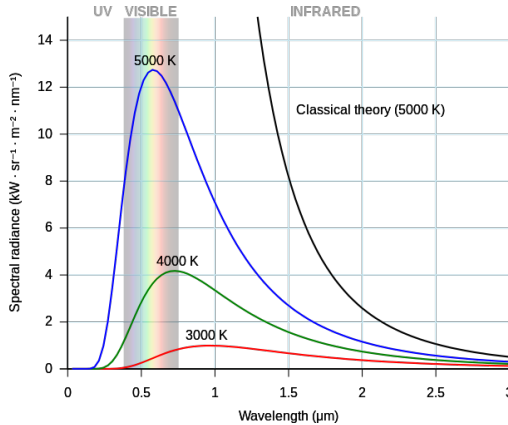
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 - A blackbody absorbs all the radiation incident on it.
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- Balfour Stewart discovered that good absorbers are good emitters.
 - His interdisciplinary work connected thermodynamics, magnetism, and meteorology, leaving a lasting impact on each field.

RADIATION

- Scientists like Rayleigh, Jeans, and Wein observed the shape of the blackbody radiation graph



THE BLACK BODY AND IT'S RADIATION

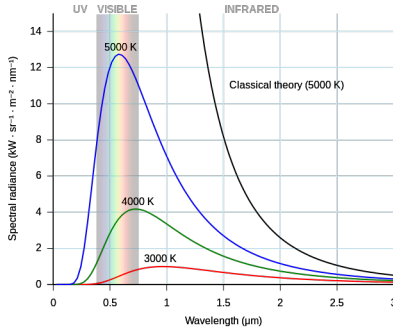


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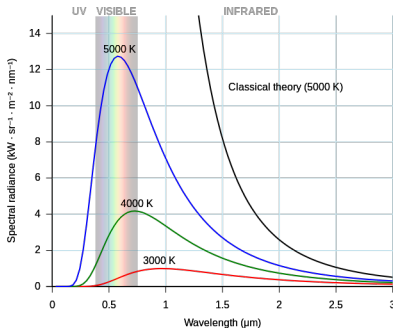
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THE BLACK BODY AND IT'S PROPERTIES



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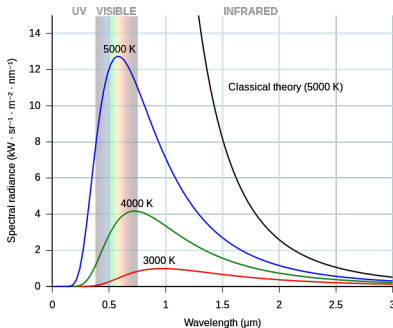


Wilhelm Wein proposed that

$$\lambda_{max} \cdot T = b$$

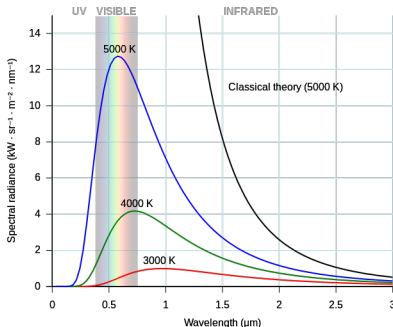
where, $b = 2.897771955 \times 10^{-3} \text{ m} \cdot \text{K}$ is the Wein's constant

THE BLACK BODY AND IT'S PROPERTIES



Knowing that the solar radiation peaks in the visible wavelengths, can you estimate the surface temperature of sun?

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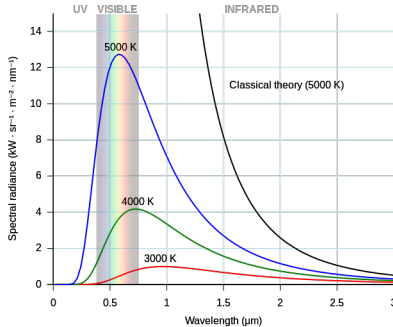


Rayleigh-Jeans proposed the classical theory of blackbody radiation

$$B_{\lambda}(T) = \frac{2ck_B T}{\lambda^4}$$

where, $k_B = 1.38 \times 10^{-23}$ J/K is the Boltzmann's constant

THE BLACK BODY AND THE CATASTROPHE



MAXWELL'S DRAMATIC PREDICTIONS

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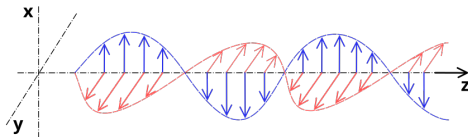
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These waves can travel through a vacuum and include a wide range of frequencies, from radio waves to gamma rays.



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UNIFICATION OF ELECTRICITY AND MAGNETISM: Maxwell's equations unified the previously separate fields of electricity and magnetism into a single, comprehensive theory of electromagnetism.

HERTZ'S MIND-BENDING DISCOVERY

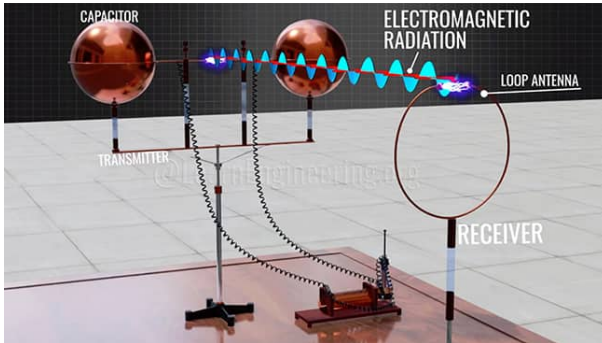


Figure: A Reconstruction of Hertz's Experiment

HERT'Z MIND-BENDING DISCOVERY

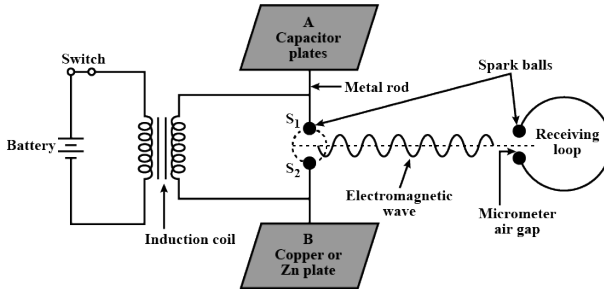


Fig : Sketch of the apparatus used by Hertz for producing and detecting radiowaves

Figure: The schematic of Hertz's Experimental Setup

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POLARIZATION: Waves could be oriented in specific directions.



HERT'Z MIND-BENDING DISCOVERY

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- 1 Measurement of Wavelength:

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HERTZ'Z MIND-BENDING DISCOVERY

1 Measurement of Wavelength:

- Hertz measured the wavelength of the waves by setting up standing wave patterns and determining the distance between nodes.
- From the wavelength and the known frequency of the spark discharge, he calculated the speed of the waves, which matched the speed of light.

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where h is Planck-Einstein-Bose constant (generally known as Planck's constant) and ν is the frequency of radiation.

PLANCK'S BLACK RADIATION LAW

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The spectral radiance $B(\nu, T)$, or energy emitted per unit surface area, per unit solid angle, per unit frequency, is given by:

$$B(\nu, T) = \frac{2h\nu^3}{c^2} \cdot \frac{1}{e^{\frac{h\nu}{k_B T}} - 1}$$

Where:

- ν : Frequency of the radiation
- T : Absolute temperature of the blackbody (in Kelvin)
- h : Planck's constant (6.626×10^{-34} Js)
- c : Speed of light (3.0×10^8 m/s)
- k_B : Boltzmann's constant (1.381×10^{-23} J/K)

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Alternatively, in terms of wavelength λ , the law is:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{e^{\frac{hc}{\lambda k_B T}} - 1}$$

Where:

- λ : Wavelength of the radiation.

WHAT THE EXPERIMENTS SHOWED US

- Existence of a Threshold Frequency

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For a given material, photoemission occurs only if the incident light has a frequency higher than a certain threshold frequency (f_0).

EXPLANATION

Classical wave theory predicted that light of any frequency, if intense enough, should cause photoemission. However, experiments showed that no electrons were emitted if the light's frequency was below f_0 , regardless of its intensity.

WHAT THE EXPERIMENTS SHOWED US

- Instantaneous Emission of Electrons

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EXPLANATION

Classical wave theory, which assumes energy is gradually absorbed by the electron, predicted a time lag before emission. The lack of a time delay indicated that the energy transfer happens in discrete packets (quanta).

WHAT THE EXPERIMENTS SHOWED US

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- The relation is linear: $K_{\max} = hf - \phi$, where h is Planck's constant, f is the light frequency, and ϕ is the material's work function.

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EXPLANATION

Classical wave theory could not explain this as it associated energy with light intensity. Einstein's theory proposed that each photon transfers its energy ($h\nu$) to a single electron.



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WHAT THE EXPERIMENTS SHOWED US

- Independence of Photoelectron Energy on Light Intensity

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EXPLANATION

According to Einstein's theory, intensity determines the number of photons (and thus the number of emitted electrons), but the energy of each photon is determined solely by its frequency ($h\nu$).

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EXPLANATION

This confirmed that the number of photons (and thus the number of electrons emitted) depends on the light intensity, while the energy of the emitted electrons is determined by the frequency.

EINSTEIN'S CONTRIBUTION

In 1905, Einstein explained these observations by proposing that light is quantized into packets of energy (photons) and formulated the photoelectric equation:

$$K_{\max} = h\nu - \phi \quad (2)$$

This equation unified the experimental observations and introduced the concept of photons, laying the foundation for quantum mechanics. It was later validated experimentally by Robert Millikan and led to Einstein receiving the Nobel Prize in Physics in 1921.

SO, WHY THE NOBEL PRIZE, THOUGH?

What was so important about the
Photoelectric Effect?

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- Confirmed the Particle Nature of Light

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THE PARTICLE THEORY OF LIGHT, REVISITED

- Confirmed the Particle Nature of Light
 - Einstein proposed that light is made of discrete packets of energy called photons, with energy $E = h\nu$, where h is Planck's constant and ν is the frequency of the light.
 - This idea challenged the classical wave theory of light, which treated light purely as a wave. By successfully explaining the photoelectric effect, Einstein provided compelling evidence for the dual nature of light (wave-particle duality).

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- Explained Observations that Classical Physics Could Not

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 - He connected the idea of light being an electromagnetic wave with frequency ν , to the idea of light as a particle with zero rest mass, but finite energy!

QUANTUM FOR THE WIN!

- Validated and expanded the Nascent Quantum Theory

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- Validated and expanded the Nascent Quantum Theory
 - Einstein's work validated Max Planck's, as yet unverified proposal of energy quantisation, by expanding its scope beyond the Blackbody Radiation.
 - His work marked a significant step in the development of quantum mechanics, which became one of the pillars of modern physics.

LIGHT POKES MATTER

- Unification of Light-Matter Interactions

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 - The equation revealed that energy transfer during the photoelectric effect occurs in discrete amounts, with each photon transferring its entire energy to a single electron.
 - This was one of the earliest and a clear mechanism for the interaction of light and matter at the quantum level.

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- It had profound theoretical and practical significance.
- It was a clear example of the triumph of the quantum theory, which became a cornerstone of modern physics.

SUMMARY

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- ➋ **Absorption Spectra:** Formed when atoms or molecules absorb specific wavelengths of light, absorption spectra appear as dark lines superimposed on a continuous spectrum.
- ➌ **Relation Between Absorption and Emission:** The wavelengths of dark lines in an absorption spectrum match the bright lines in the corresponding emission spectrum of the same element, reflecting the quantized energy levels of atoms.

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Emission and Absorption Spectra

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- ➍ **Fraunhofer Lines:** These are dark absorption lines observed in the Sun's spectrum caused by elements in the Sun's cooler outer layers absorbing specific wavelengths from the continuous spectrum produced by the hotter interior.
- ➎ **Energy Quantization:** The specific lines correspond to discrete energy differences between quantized electron energy levels.

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- ❹ **Applications:** Blackbody radiation concepts are applied in thermal imaging, astrophysics (cosmic microwave background), and determining stellar temperatures.

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- 2 **Einstein's Explanation:** Albert Einstein proposed that light consists of photons with energy $E = h\nu$. An electron is ejected if the photon's energy exceeds the material's work function (ϕ), demonstrating the particle-like behavior of light and earning Einstein the Nobel Prize in 1921.

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